Mahesh Chitakoti

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Summary

Full Stack & Applied AI Engineer with 3+ years of software engineering experience building web applications and LLM-powered features with **React.js, Next.js, TypeScript, Node.js, and FastAPI**. Increased signup conversion by **18%**, reduced median API latency by **60%**, and cut recurring LLM inference costs by **35%**.

Technical Skills

Frontend Engineering: React.js, Next.js, TypeScript, JavaScript (ES6+), Redux Toolkit, Tailwind CSS, Vite

Backend & APIs: Node.js, Express.js, Python, FastAPI, asyncio, REST APIs, WebSockets, Microservices, Idempotent Webhooks

Databases & Caching: PostgreSQL, Redis (Caching, Task Queues), Compound Indexing, Connection Pooling

Cloud, DevOps & Quality: AWS (EC2, S3), Docker, GitHub Actions CI/CD, Jest, Postman API Testing, Linux, Nginx

Applied AI: RAG, Agentic Workflows (LangGraph), Qdrant, pgvector, Semantic Caching, Structured Outputs (Pydantic), Gemini & OpenAI APIs

Experience

Full Stack & AI Engineer (Contract) – PearlThoughts May 2026 – Present

Client Projects: Document Processing & Search *Remote*

- Implemented a **FastAPI** backend that handled client webhooks and used **Redis queues** to schedule data extraction and schema validation for multi-page PDF invoices.
- Deployed containerized services with **Docker on AWS EC2**, provisioned **S3** object storage, and automated linting and tests with **GitHub Actions**.
- Reduced average vector retrieval latency from **280 ms to under 185 ms** using **PostgreSQL with pgvector HNSW indexing** and semantic caching.

AI Developer – SahioNext Technology Pvt. Ltd. Feb 2025 – Apr 2026

Product: Tulkka, an AI-powered English learning platform *Remote*

- Developed **React.js** interfaces and **FastAPI** services for Tulkka's AI-powered audio evaluation. Used **PostgreSQL** for storage and **Redis queues** to dispatch audio processing jobs to workers.
- Built **React.js and Redux Toolkit** dashboards that surfaced token usage, rate limits, and pipeline bottlenecks, reducing time to triage pipeline incidents from hours to minutes.
- Implemented full-duplex **WebSocket endpoints** for real-time speech evaluation, using client-side buffering and automatic reconnection to recover sessions on unstable networks.
- Integrated **Qdrant hybrid search** into the evaluation workflow to retrieve relevant content from **10,000+ lesson transcript chunks** at p95 latency under **300 ms**.
- Cut recurring LLM inference costs by **35%** using exact-match caching and **Redis-backed semantic caching**.

Full Stack Developer – The DevOps Team Oct 2023 – Jan 2025

Product: Recruiting Platform *Bengaluru, India*

- Built responsive onboarding and profile workflows in **React.js and Next.js**. A/B-tested changes to the multi-step signup form increased registration conversion by **18%**.
- Optimized **Node.js and Express.js APIs** using Redis caching, PostgreSQL compound indexes, and connection pooling, reducing median response times from **800 ms to 320 ms**.
- Added **Jest** unit tests and **Postman** API tests for core authentication and search endpoints.

Projects

WorkBridge – Project Collaboration Platform *Sep 2026 – Present*

Next.js, *React.js*, *TypeScript*, *Node.js*, *Express.js*, *FastAPI*, *PostgreSQL*, *pgvector*, *Tailwind CSS* | *Live Project*

- Built **Next.js and React.js** interfaces for project discovery and submissions, giving founders and hiring teams a way to evaluate developers through project work and GitHub repositories.
- Implemented a **Node.js/Express API gateway** for GitHub OAuth 2.0, session handling, and role-based access control.
- Integrated a **FastAPI** service that uses LLMs to assess code quality in submitted GitHub repositories. Used **PostgreSQL (pgvector)** semantic search to match developers by relevant project experience.

Certifications

Oracle Agentic AI Certified Foundations Associate – Oracle — Aug 2026
Oracle AI Vector Search Certified Professional – Oracle — Sep 2025
Machine Learning I – Columbia University — Aug 2025
Oracle Cloud Infrastructure Generative AI Professional – Oracle — Aug 2024

Education

Adichunchanagiri Institute of Technology (VTU) — Chikkamagaluru, KA, India
Bachelor of Engineering (B.E.) — *Aug 2023*